Clinical Nutrition ESPEN 46 (2021) 386–393

Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: http://www.clinicalnutritionespen.com

Original article

Comparison between handheld ultrasound and regional and whole-body dual energy x-ray absorptiometry (DXA) for body fat assessment

Sandra L. Gomez-Perez [a, *], Yanyu Zhang [b], Marina Mourtzakis [c], Lisa Tussing-Humphreys [d], Jason Ridlon [e, f], H. Rex Gaskins [e, f], Ece Mutlu [g]

[a] Rush University, Department of Clinical Nutrition, Chicago, IL, USA
[b] Rush University Medical Center, Rush Bioinformatics and Biostatistics Core, Chicago, IL, USA
[c] University of Waterloo, Department of Kinesiology, Waterloo, Ontario, Canada
[d] University of Illinois at Chicago, Department of Kinesiology and Nutrition, UIC Cancer Center, Chicago, IL, USA
[e] University of Illinois Urbana-Champaign, Department of Animal Sciences, Urbana-Champaign, IL, USA
[f] Cancer Center at Illinois, Urbana-Champaign, IL, USA
[g] Rush University Medical Center, Department of Medicine, Chicago, IL, USA

ARTICLE INFO

Article history:
Received 25 August 2021
Accepted 30 August 2021

Keywords:
Body fat assessment
Body composition
Abdominal adiposity
Ultrasound
DXA

ABSTRACT

Objectives: To determine the extent of agreement between a handheld ultrasound (US) attached to an android tablet and the reference method dual energy x-ray absorptiometry (DXA) for the measurement of adiposity.

Methods: A whole-body DXA scan and abdominal adipose tissue thickness measurements using a handheld US were obtained from 104 adults (63 females, 41 males). Body fat percent (BF%), total fat mass (kg), and trunk fat mass (kg) were obtained from DXA. Subcutaneous adipose tissue (SAT), superficial subcutaneous adipose tissue (SSAT), and deep subcutaneous adipose tissue (DSAT) thickness were obtained from US. Sex-specific total fat mass, trunk fat mass, and BF% estimates by US were compared with DXA. Spearman's correlations and Bland–Altman plots were used to assess agreement between the methods.

Results: US SAT correlated strongly with total fat mass for both females ($r_s = 0.74$) and males ($r_s = 0.87$) as did trunk fat mass (females, $r_s = 0.81$; males, $r_s = 0.83$); as did SSAT and DSAT (females: $r_s = 0.65$ and $r_s = 0.66$; males: $r_s = 0.63$ and $r_s = 0.85$, respectively, all p-values < 0.0001). Bland–Altman plots demonstrated strong agreement for total and trunk fat mass for both males and females. For BF%, acceptable limits of agreement were observed for males but not for females, substantial proportional bias as indicated by a negative slope was noted for BF% using SAT ($r = -0.298$, $p = 0.0177$).

Conclusion: The handheld US and technique to analyze abdominal adipose tissue thickness showed strong agreement with DXA results and generated highly comparable estimates for total and trunk fat mass for both sexes.

© 2021 European Society for Clinical Nutrition and Metabolism. Published by Elsevier Ltd. All rights reserved.

1. Introduction

Obesity is a particularly challenging public health issue of multifactorial etiology with significant heterogeneous body composition phenotypes [1,2]. Although body mass index (BMI) has been used extensively to identify health risks associated with obesity at the population level, it is an imprecise measure of body composition and fails to differentiate patterning and distribution of both adipose and muscle at the individual level [3,4]. Undesirable changes in body composition including excess visceral adipose

* Corresponding author. Rush University Medical Center, Department of Clinical Nutrition, Armour Academic Center, 600 S. Paulina St, AAC 737D, Chicago, IL, 60612, USA.
E-mail address: sandra_l_gomez-perez@rush.edu (S.L. Gomez-Perez).

https://doi.org/10.1016/j.clnesp.2021.08.038
2405-4577/© 2021 European Society for Clinical Nutrition and Metabolism. Published by Elsevier Ltd. All rights reserved.

tissue (VAT) accumulation in response to weight gain have been postulated as significant contributors to "metabolic unhealthy" obesity however, specific mechanisms have yet to be revealed [2,5–7]. Additionally, emerging interest in subcutaneous adipose tissue (SAT) and its subtypes, superficial subcutaneous adipose tissue (SSAT), and deep subcutaneous adipose tissue (DSAT) have indicated that, DSAT in particular, may also contribute significantly to metabolic dysfunction, evidenced by studies assessing the relationship of the latter two adipose depots with cardiometabolic risk in healthy adults and patients with type 2 diabetes [8–10]. Thus, it has become increasingly important not only to identify the presence of obesity with measures like BMI, but also to quantify distribution of excess adipose tissue by body composition assessments like dual energy x-ray absorptiometry (DXA), computed tomography (CT) and ultrasound (US) that provide direct estimates of adipose and muscle tissue distribution, and thereby increase the predictive potential of metabolic disease risk compared to BMI alone [11–15].

Ultrasound has a distinct advantage to study body composition in masses: It carries no risk of physical harm; can be easily performed at the bedside using a small, handheld, and cheap probe that can fit into a pocket (which nowadays gives similar results to that of large US machines); and can be completed and interpreted in minutes. Alternatives (which are the current gold standards) such as DXA and MRI require specialized space and expensive equipment; trained staff; take longer to perform and interpret; are more expensive; carry a risk of radiation exposure (albeit small) or cannot be performed in certain people (e.g. those with pacemakers, metal in their bodies, claustrophobia, etc.). In fact, many medical students have begun training in US, which is expected to be widely utilized within the next decade at the bedside. Furthermore, developments in image processing with AI are also expected to enhance obtaining results very rapidly in real-time. As a research tool, US has the advantage of minimal cost, minimal risk, high ease-of-use, and high portability to anywhere and everywhere in and out of hospital-based medical settings. Such ease, cost, and practicality of US represents an unprecedented opportunity for use in body composition assessment, especially in the community at a large scale (which is not feasible with DXA or MRI), or in various outpatient and inpatient clinical settings without having to send a patient to a radiology department or facility.

Historically, US has been used in body composition research for over 30 years using large sophisticated hospital-based ultrasound equipment, however, there is a lack of consensus on the best US technique for this purpose [13,16,17]. Recently, several excellent reviews explaining the advantages and disadvantages of various body composition methods, including ultrasound have been published [16,17]. These reviews state that the US is non-invasive, quick, valid and reliable, and relative inexpensive. Yet, much research is needed to establish US as wide-spread technique for the evaluation of body composition. To date, there has not been definition of a practical, best-practice, bedside US protocol with standardized anatomical landmark(s) that yield(s) reasonable and reproducible estimates of total body adiposity and skeletal muscle mass. Various researchers have recently advanced US science through validations of 9-site and 5-site US techniques, to estimate whole-body skeletal muscle mass and body fat mass against DXA [18,19]. However, multi-site US techniques are impractical, as they are time-intensive and more burdensome and less pragmatic for a wider range of clinical and non-clinical populations, compared to single-site US techniques [16,17]. Validation studies have been published using US for single-anatomical sites of the forearm muscle and abdominal region (5 cm left and right of the umbilicus) compared to reference methods, DXA or MRI, respectively [9,20,21]. These techniques with proper training and calibration show promise for estimating both total body fat and muscle using portable US devices. Marinou et al. measured abdominal fat thicknesses using US (Philips PDI 5000) for SAT and its two layers, SSAT and DSAT. They reported this method to be strongly correlated with the measurements taken by MRI [9]. The latter technique by Marinou et al. demonstrates great promise for abdominal adipose tissue assessment; and requires further testing and validation in different settings and populations; and especially using a portable inexpensive device. With adequate and consistent validation, we propose that measurement of abdominal adipose thickness at a single-site by US may prove useful for the prediction of total fat mass, % body fat (BF%) and trunk fat mass in large population-based studies, specifically targeting obesity prevention and management.

Thus, for this first study we utilized a portable, cheap and easy-to-use handheld US device attached to an android tablet (Fig. 1) that can be deployed in numerous settings including medical management, clinical research, community – based research and large epidemiological studies, and evaluated the agreement of a single-site abdominal US technique published by Marinou et al. [9] for measuring SAT, SSAT, and DSAT thicknesses against the gold-standard DXA criterion method to estimate different parameters of regional and total fat mass in relatively healthy middle-aged adults.

2. Subjects and methods

Data were collected on a sample of 104 healthy adults (63 females, 41 males) participating in a multi-institutional study (*Diet modulation of bacterial sulfur and bile acid metabolism and colon cancer risk* [R01CA204808-01]; Rush University Medical Center Protocol No.: 13102201; University of Illinois at Chicago Protocol No.: 2016-0495). Participants were recruited from outpatient endoscopy clinics from patients scheduled for a clinically indicated screening or surveillance colonoscopy. Participants were included if they were African American or White adults; additionally, females had to be post-menopausal defined as no menstruation for the past 6 months. Participants were excluded for the following reasons: 1) race other than White or Black/African American, 2) current cancer 3) current gastrointestinal illness including inflammatory bowel disease, celiac disease, 4) history of chronic kidney or cardiac disease, 5) history of positive genetic test for inherited polyposis syndromes such as familial adenomatous polyps, 6) history of alcohol or illicit drug use, 7) antibiotic use in the previous 2 months, 8) use of medications that interfere with normal digestion, 9) pre or pro-biotic use, 10) anticoagulant use, 9) non-English speaking, or 10) pregnant or breastfeeding women. Body composition assessment using DXA and portable handheld US were performed (Fig. 1) on each participant. Anthropometric, demographic and clinical information including DXA and US were collected at a research visit approximately 4 weeks after the clinical procedure. All participants signed a written informed consent prior to study participation.

2.1. Anthropometric assessment

Anthropometric measurements were performed by trained research assistants on same day of DXA and US measurements. Height was measured twice and recorded in centimeters (cm) using a portable stadiometer (Seca, Germany), without shoes. Weight was measured twice, in light clothing and after pockets were emptied, and recorded in kilograms (kg) using a digital scale (Tanita BWB-800, Arlington Heights, IL). Body mass index (BMI) was then calculated using the following formula: weight (kg)/height (m)^2.

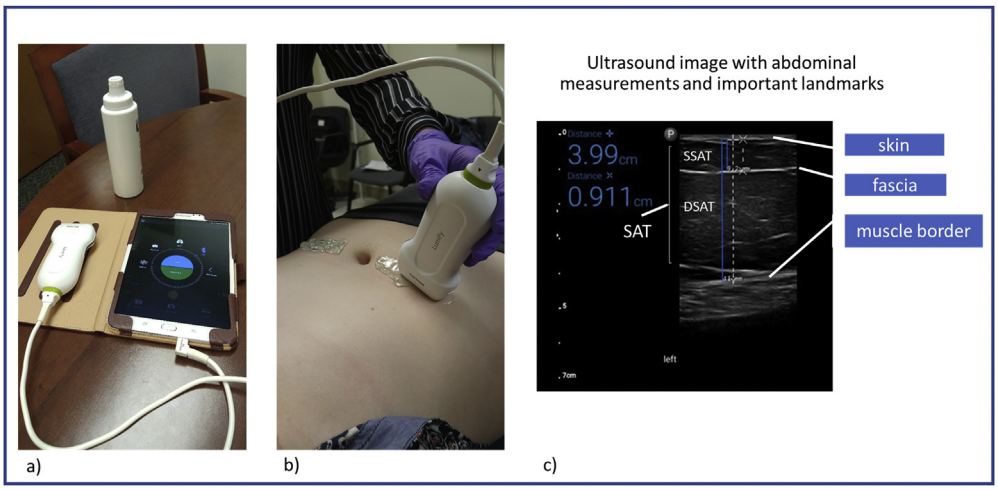

Fig. 1. Portable, handheld ultrasound attached to android tablet (a) used in this study. A subject being scanned with ultrasound (b) and (c) an image with abdominal landmarks showing thickness measurements for subcutaneous adipose tissue (SAT) and subtypes: superficial subcutaneous adipose tissue (SSAT) and deep subcutaneous adipose tissue (DSAT).

2.2. DXA assessment

DXA is a criterion method for human body composition. It has the capacity to provide highly accurate estimates of whole-body fat mass and fat-free mass as well as regional analyses and appendicular (limbs) skeletal muscle mass [12]. A whole-body DXA scan was performed using the GE iLunar (GE Healthcare, Madison, WI) by a trained researcher. DXA was calibrated daily using the GE Lunar block calibration phantom to confirm that the scanner was performing according to the specifications dictated by the manufacturer. The scans were analyzed by a single trained researcher using DXA software version: enCORE version 17 © 1998–2016 (GE Healthcare, Madison, WI). Participants wore light, non-metal clothing, had eaten a light snack 30–45 min and voided immediately before the scan. All metal objects including glasses, jewelry, and wallets were collected prior to the procedure. The DXA body composition parameters recorded were total fat mass (kg), BF% and trunk fat mass (kg). Total fat mass (whole-body), BF% and trunk fat (neck, trunk and pelvis) mass estimates were defined according to default settings of enCORE DXA software.

2.3. Ultrasound assessment

The abdominal US was conducted using the Philips Lumify (Philips Ultrasound, Bothall, WA) handheld portable B-mode L12-4 linear array transducer that attaches to an android tablet and works through the Lumify App (Software Version 45351940111_1.7.1.49 [2017/09/25]-05:50) (Fig. 1). The abdominal US measurements were taken 5 cm horizontally on the right and left side of the abdominal wall at the umbilicus in the supine position (on DXA scanner), using minimal compression as previously published [9]. Two thickness measurements recorded in cm were taken from each side of the umbilicus at the end of quiet exhalation and averaged for SAT, SSAT, and DSAT measurements.

Two raters were trained to perform US measurements however, 96% of the measurements were performed by only one of the raters. The second rater was trained to function as back-up and performed US measurements only when needed. Five research volunteers were measured by the two trained raters for inter-rater reliability. For each volunteer, each rater performed the ultrasound technique taking two measurements from both sides of the umbilicus. There was a 5-min break between raters. Minimal compression was used during measurements. Landmarking and ultrasound gel on the abdominal area were removed between raters. Measurements for each of the five volunteers were recorded live using the US application as the measurements were being taken. Intra-rater reliability was determined from the measurements taken by the same rater for each volunteer measured. Intra-rater and inter-rater reliability were performed using the intraclass correlation coefficient (ICC).

2.4. Statistical analysis

All statistics were conducted using SAS 9.3 (SAS Institute Inc.). Frequency distributions were used for categorical variables and mean (standard deviation [SD]) was used for continuous variables. Demographic and anthropometric measurements (height, weight, BMI) were calculated for the total sample and then stratified by sex. The percentage of adults with normal, overweight and obese BMI were calculated using classifications as previously defined [22]. Spearman's correlation (r_s) was performed between DXA and US measurements to assess associations between the raw estimates of abdominal thickness using US for SAT (cm), SSAT (cm), and DSAT (cm) against DXA values for total body fat mass (kg), BF% and trunk fat (kg). Multivariate linear regression equations were used to estimate total fat mass, BF% and trunk fat from abdominal US thickness measurements (SAT) against the DXA reference method estimates for total body fat, BF%, and trunk fat. Agreement for Spearman's (r_s) correlations were defined as follows: <0 no agreement, 0–0.20 very low (slight), 0.21–0.40 fair (low), 0.41–0.60 moderate, 0.61–0.80 as substantial (high), and 0.81–1 almost perfect or very high [16,23]. Multivariate regression models were used for US include BMI and SAT thickness estimates. The dependent variables were DXA derived total fat mass, trunk fat mass, and BF% and the independent variables were US_SAT, US_SSAT and US_DSAT. The covariates of age, weight and BMI were included in regression models only if they significantly predicted dependent variables or improved the models. Age was not a significant predictor of dependent variables (See Supplementary Materials) and weight was inferior in model fit to BMI and thus excluded. Most parsimonious models were ultimately selected. Multicollinearity defined as a variance inflation factor >10 was not observed for any of the regression models (VIF range 1.57–3.41). To further understand the estimated US values from regression equations in this adult population, the adjusted R^2 of multivariate

regression models between the reference method, DXA, and estimated adiposity values (total fat mass, BF%, and trunk fat) from the US and standard errors of these estimates (SEE) were also evaluated, as they have been previously described in other papers [12,24]. Body composition parameters for the reference method DXA and for the US method were estimated for the total sample and then according to sex. Wilcoxon rank sum tests for non-normally distributed continuous variables were used to compare total fat mass, BF%, and trunk fat estimates from US and DXA for total sample and then by sex. Bland–Altman plots were used to estimate the 95% limits of agreement between DXA reference estimates compared to regression equations using US data [25,26]. The 95% limits of agreement range was calculated by: Mean ± 1.96 × SD, where mean is the average of the difference between DXA and estimated quantities by US regression equation; SD is the standard deviation of the difference between DXA and estimated quantities by US regression equation. The existence of proportional bias was tested by evaluating the slope of the regression line fitted to the Bland–Altman plot. The percent errors ((DXA estimate − US derived estimate)/(DXA Estimate) were calculated to determine the percentage of subjects exhibiting a difference between the techniques below 5% (acceptable percent error between methods). Significance was set at $p < 0.05$.

2.5. Sample size estimates

Using an effect sizes of 0.70 and 0.80 for our three outcomes (with both male and females, US SAT, US SSAT, and US DSAT), and an alpha = 0.05, power = 80%, the total sample sizes required were 13 and 9, respectively. Calculations were performed in R with package "pwr" for correlation coefficients. Given that the sample size is often determined based on the first aim of the study, we also conducted a *post hoc* sample size estimation using the lowest correlation coefficient between trunk fat mass (DXA body composition parameter) and SSAT thickness (US body composition parameter) ($r_s = 0.48$). Using G*Power [27], with this effect size ($r_s = 0.48$), an alpha = 0.05, power = 0.80, the total sample needed equals to 31. Thus, using the most conservative estimate of 31 per gender group, we were sufficiently powered to develop equations for both males and females.

3. Results

3.1. Sample demographics and anthropometric characteristics

A total of 104 relatively healthy adult males and females were included in the analysis. Table 1 contains a summary of the subject characteristics including demographic, anthropometric and body composition data. The median (IQR) for the total sample for age was 60 (+/− 6) years and the median BMI was 31 (+/−7) kg/m^2. Our sample was mostly obese (50%), followed by overweight (28%) and normal BMI (22%) subjects. Approximately, 49% (31/63) of women and 51% (21/41) of men in our sample were overweight or obese. The sample was more female (61%) and 53% were Black/African Americans. The mean (+/-SD) for SAT thickness was higher for females than males (3.3 ± 1.3 cm vs 2.4 ± 1.2, respectively).

Correlations between DXA measurements, total fat mass, BF% and trunk fat mass and US thickness estimates for SAT were 0.79, 0.75, and 0.75; for SSAT were 0.59, 0.7 and 0.48; and for DSAT were 0.76, 0.61 and 0.77, respectively. All correlations between the two methods were significant and positively related. The ICC for intra-rater reliability was 0.96 and for inter-rater reliability was 0.98 for US measurements.

3.2. Ultrasound regression equations for DXA reference estimates of body composition parameters

The regression equations to estimate total fat mass (kg), BF % and trunk fat mass (kg) using US SAT, US DSAT and US SSAT thickness data for females and males are found in Table 2.

The estimates obtained from the US for total fat mass, trunk fat mass and BF% were not different for females or males when compared to reference estimates by DXA ($p > 0.05$). The determined estimates for US for trunk fat mass were not different from DXA values for females or males ($p = 0.81$ and $p = 0.78$). For trunk fat mass, BF% derived by US was strongly related to the DXA estimate for both females and males (Adjusted $R^2 \geq 0.86$, $p < 0.01$). Additionally, as seen in Table 3, the SAT estimates from the US compared to DXA (36 ± 13 vs. 36 ± 13 kg, respectively) for total fat mass in females was not different ($p = 0.76$). The SAT estimates were also very similar between US and DXA for the males ($p = 0.62$).

3.3. Agreement between DXA and US measurements

Bland Altman plots (Fig. 2) revealed that US SAT estimates of total fat mass (females ± 6.87; males ± 9.05), BF% (females ± 7.11; males ± 7.33), and trunk fat mass (females ± 5.46; males ± 6.43) fell within the limits of agreement (95% CI) for both females (±6.87) and males (±9.05). No significant proportional bias was observed between US and DXA for total fat mass for females ($r = -0.13$, $p = 0.29$), total fat for males ($r = -0.16$, $p = 0.32$), BF% for males ($r = 0.24$, $p = 0.12$), trunk fat mass for females ($r = -0.18$, $p = 0.16$) and trunk fat mass for males ($r = -0.18$, $p = 0.25$). However, a significant negative slope was associated with BF% for females ($r = -0.30$, $p = 0.02$) which indicates a notable proportional bias.

Table 1
Sample characteristics.

Characteristics	Total (N = 104)	Females (n = 63, 61%)	Males (n = 41, 38%)
Age, years	60.40 (6.25)	59.71 (6.33)	61.46 (6.05)
Ethnicity, n (%)			
Black African American	55 (53)	37 (59)	18 (44)
Non-Hispanic White	49 (47)	26 (41)	23 (56)
Weight, kg	92.97 (27.06)	87.13 (27.26)	101.93 (24.45)
Height, cm	170.06 (9.51)	164.51 (6.07)	178.58 (7.26)
BMI, kg/m^2	31.00 (6.94)	30.86 (7.18)	31.22 (6.63)
US abdominal fat thickness, cm			
SAT (left + right)/2	2.96 (1.35)	3.29 (1.35)	2.45 (1.17)
SSAT (left + right)/2	1.24 (0.66)	1.53 (0.65)	0.80 (0.37)
DSAT (left + right)/2	1.72 (0.91)	1.77 (0.92)	1.64 (0.91)

Data presented as mean (SD) unless otherwise specified. BMI = body mass index; US = ultrasound; SAT = subcutaneous adipose tissue; SSAT = superficial SAT; DSAT = deep SAT.

Table 2
Ultrasound regression equations for DXA reference estimates of body composition parameters.

DXA-derived Body composition parameter	Ultrasound thickness parameter	Gender	Regression Equations	R-squared	Adjusted R-squared	SEE
Total fat mass (kg)	SAT	Males	= −26.13 + (1.62*BMI) + (3.28*US SAT)	0.90	0.90	2.23
		Females	= −18.09 + (1.66*BMI) + (0.91*US SAT)	0.93	0.93	1.92
	SSAT	Males	= −32.11 + (1.97*BMI) + (3.82*US SSAT)	0.89	0.88	2.31
		Females	= −18.87 + (1.74*BMI) + (0.80*US SSAT)	0.93	0.93	1.94
	DSAT	Males	= −26.20 + (1.68*BMI) + (3.81*US DSAT)	0.90	0.90	2.24
		Females	= −18.01 + (1.69*BMI) + (1.17*US DSAT)	0.93	0.93	1.92
Body fat percentage (%)	SAT	Males	= 6.19 + (0.59*BMI) + (3.26*US SAT)	0.80	0.79	2.01
		Females	= 19.16 + (0.74*BMI) + (0.50*US SAT)	0.72	0.71	1.98
	SSAT	Males	= - 0.13 + (0.98*BMI) + (2.64*US SSAT)	0.74	0.73	2.13
		Females	= 18.74 + (0.78*BMI) + (0.49*US SSAT)	0.72	0.71	1.96
	DSAT	Males	= 6.73 + (0.61*BMI) + (4.11*US DSAT)	0.80	0.79	1.99
		Females	= 19.18 + (0.75*BMI) + (0.61*US DSAT)	0.72	0.71	1.95
Trunk fat mass (kg)	SAT	Males	= −19.29 + (1.16*BMI) + (1.06*US SAT)	0.88	0.88	0.39
		Females	= −12.25 + (0.83*BMI) + (1.53*US SAT)	0.88	0.87	0.83
	SSAT	Males	= - 21.23 + (1.27*BMI) + (1.21*US SSAT)	0.87	0.86	1.95
		Females	= −13.51 + (0.96*BMI) + (1.74*US SSAT)	0.87	0.86	0.75
	DSAT	Males	= −19.30 + (1.18*BMI) + (1.24*US DSAT)	0.88	0.87	1.02
		Females	= −12.30 + (0.90*BMI) + (1.75*US DSAT)	0.88	0.87	0.54

Abbreviations: SAT (subcutaneous adipose tissue), SSAT (superficial subcutaneous adipose tissue), DSAT (deep subcutaneous adipose tissue), SEE (standard error of the estimate). Males: n = 41; Females: n = 63.

Table 3
Body composition parameters using DXA and US.

Characteristics	Total (N = 104)[c]	Females[c] (n = 63, 61%)	Males[c] (n = 41, 38%)
DXA body fat[a]			
Total fat mass, kg	35 (14)[a]	36 (13)[d]	33 (15)[g]
Body fat %	39 (9)[b]	43 (7)[e]	33 (8)[h]
Trunk fat mass, kg	29 (9)[c]	19 (8)[f]	20 (9)[i]
US body fat[b]			
Total fat mass, kg	35 (13)[a]	36 (13)[d]	33 (14)[g]
Body fat %	39 (7)[b]	43 (6)[e]	33 (7)[h]
Trunk fat mass, kg	19 (8)[c]	19 (8)[f]	20 (9)[i]

DXA = dual energy x-ray absorptiometry; US = ultrasound. Body fat % range: 13.2–59.4%; total fat mass range = 7.97–68.78 kg; trunk fat mass range = 1.71–42.94 kg.
[a] Reference method.
[b] Estimated from multivariable regression equations including US SAT thickness values and BMI.
[c] Wilcoxon sign rank tests were used to compare US vs DXA estimates for total sample for total body fat mass (p = 0.76[a]), body fat % (p = 0.89[b]), trunk fat mass (p = 0.73)[c]; for females for total body fat mass (p = 0.75)[d], body fat % (p = 0.50)[e] and trunk fat mass (p = 0.81)[f]; and for males for total body fat mass (p = 0.62)[g], body fat % (p = 0.84)[h], and trunk fat mass (p = 0.77)[i]; *p-value < 0.05.

Bland Altman plots (Supplementary Material) revealed that US SSAT and DSAT estimates for total fat mass, BF% and trunk fat mass fell within limits of agreement for both genders.

Outliers (>2 SD) were determined for SAT thickness and BF% to further explore the increased proportional bias observed in women, especially women with higher amounts of BF. The lower and upper limits were 31.93% and 55.17% for BF% and 0.59 and 6.0 cm for SAT thickness, respectively. The mean ± standard deviation (SD) of BF% for the total sample of women (n = 63) was 43.55 ± 5.81% and for SAT thickness was 3.30 ± 1.35 cm. The mean BMI and age were 30.86 kg/m^2 and 59.71 years, respectively. A total of 3 women had BF% above the upper limit for outliers (>55.17%). For this group of women, the mean ± SD for age, BMI, BF%, and SAT were 53.67 years, 48.70 kg/m^2, 57.70%, and 5.29 cm, respectively. The percentage of subjects that exhibited a percent error below 5% were 69.8% and 68.3% for total fat mass, 90% and 73% for BF%, and 69.8% for trunk fat mass for women and men, respectively.

4. Discussion

This is the first study, to our knowledge that utilizes a handheld portable US probe attached to an android tablet for body composition assessment. When compared to DXA, the US derived estimates for total fat mass, BF% and trunk fat mass indicated strong agreement as confirmed by Bland–Altman plots and proportional bias assessments. The three US-derived estimates for males also showed substantial agreement with DXA values. For females, the US derived estimates for total fat mass and trunk fat mass showed substantial agreement but BF% did not. This study is important because it demonstrates that a single-site abdominal US was useful for the prediction of total fat mass, BF% and trunk fat mass in adults. Thus, this technique using an ultra-portable US is practical to use in large population-based studies specifically targeting obesity prevention and management.

Our study confirms that the abdominal US technique published by Marinou et al. performs exceptionally against DXA, with highly accurate and valid SAT thickness estimates ($r_s = 0.79$, $p < 0.0001$) and aligns with their findings using MRI as the reference method ($r = 0.85$, $p < 0.0001$) [9]. Furthermore, the SAT abdominal layer thickness estimates obtained from the abdominal US technique can be placed into the sex-specific prediction equations provided by our study to generate truncal and whole-body total fat estimates for males and females in an expedient manner. Because US is less invasive, faster, harmless, cost-effective, and more portable than DXA [15], this technique and the US-derived prediction equations can be effectively applied to a larger sample of patients inside and outside of clinical settings. With further validation, the abdominal adipose tissue estimates obtained from these prediction equations can be used to test associations of total adiposity with health-related outcomes, to track changes in abdominal adiposity in both clinical and non-clinical settings, and to determine the effectiveness of interventions that focus on promoting healthy body weight and body fat distribution. Currently, there are 6 actively enrolling studies testing the US in the evaluation of sarcopenia, or low skeletal muscle, in clinical populations according to ClinicalTrials.gov suggesting that there is great interest in applying this device in clinical settings to measure and monitor body

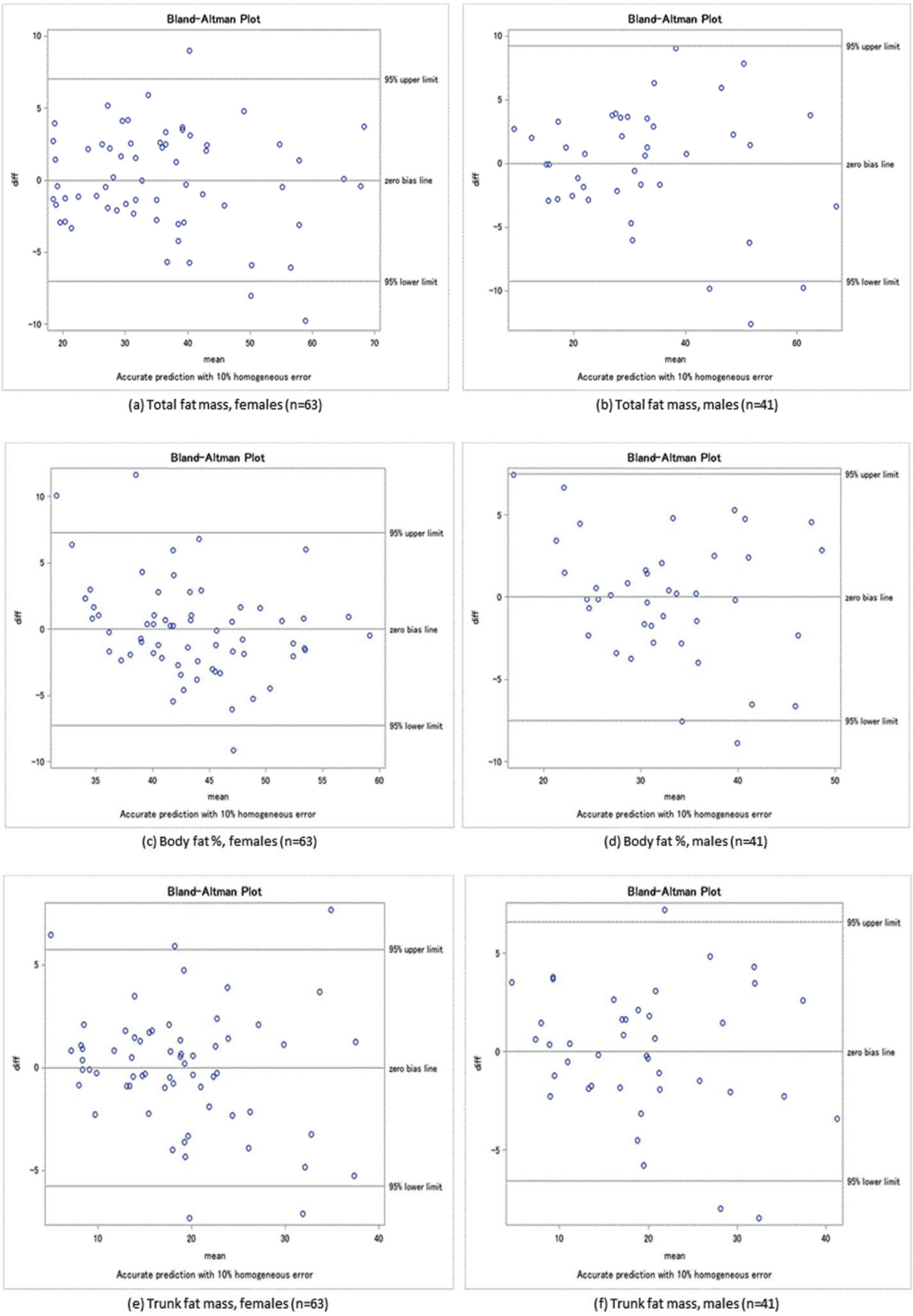

Fig. 2. Bland–Altman plots comparing total fat mass (a, b), body fat % (c, d), and trunk fat mass (e, f) by DXA and predicted total fat mass, body fat %, and trunk fat mass by US for females and males.

composition. We anticipate that the US will gain greater popularity for body composition assessment following consensus on US protocols that standardize landmarks and techniques in clinical and non-clinical (population-based) studies.

Our results support those of various other studies that have demonstrated agreement and strong associations of US-derived adipose tissue thickness estimates using multi-site (such as upright with posterior landmarks such as the subscapular or posterior upper arm; posterior abdominal wall and midthigh) protocols compared to measures of adiposity from MRI, air displacement plethysmography, bioimpedance analysis, hydrostatic weighing, and DXA, in various populations (i.e., athletes, healthy mid-age adults, older adults) [28–33]. Given its portability, convenience and safety, the data suggests that hand-held portable US has the potential to be the method of choice to assess body composition in healthy, free-living community and clinical populations and especially pragmatic for research at a low cost, allowing generation of data at the bedside.

This study has several limitations. An important limitation is that regression equations were not validated in a separate cohort. This means that our results should be interpreted as pilot conclusions that lays the groundwork for more definitive study. As such, we are currently conducting a separate validation study of these regression equations. Further testing of these equations in different cohorts than ours is also needed for independent validation. The use of landmarking at the umbilicus for the abdominal US technique may be limited due horizontal fat folds hiding the umbilicus for abdominally obese individuals: this was observed in two patients who were excluded from this analysis. Although DXA is a reference method for body composition that provides whole body and regional fat mass estimates, it does not differentiate distinct types of adipose tissue depots like visceral and intermuscular. A computed tomography (CT) or MRI could have been ideal reference methods to compare to the US, since these two methods can provide direct estimates of thickness of each of the specific adipose depots that were measured by US in this study. However, CT imaging is not ethical to be prospectively performed outside of the clinical setting and not appropriate for research due to ionizing radiation exposure. Similarly, MRI imaging is expensive and not as accessible or readily available for research purposes as DXA. Thus, it is important to acknowledge that whole body or regional fat mass estimates from the DXA do not provide information about other adipose tissues that may be contributing to the whole body or regional estimates beyond SAT and its subtypes. Additionally, our sample was mostly female and represented an older (~60 years) population of mostly obese adults from two major race/ethnic groups within the United States, thus these prediction equations may or may not be applicable or suitable for other populations. Despite that the sample is mostly female, the males performed similarly. Lastly, in women, BF%, produced a high degree of proportional bias. However, the other adiposity measures were in good agreement except BF% which includes measures of weight and may increase the error as weight includes muscle mass not just fat mass in the calculation.

We focused on total SAT and its layers, SSAT and DSAT in our study, however, the technique used in this study described originally by Marinou et al. may be effective in predicting regional and total adiposity as well as skeletal muscle. Moreover, future studies are needed to determine whether thickness estimates at the abdominal landmark either alone or when combined with other landmarks will also yield valuable information not only of tissue quantity but also 'quality' or degree of fatty infiltration of skeletal and adipose tissues and relating these measures back to DXA or other reference methods [13,34]. Degree of fatty infiltration is an indicator of tissue health and function and some researchers have been evaluating the US's echogenicity feature as a prognostic tool to measure this [13]. Future studies are needed that further examine the US estimates relative to android and gynoid fat estimates from DXA.

5. Conclusion

Herein, we describe the use of a portable, handheld US probe and an android tablet at the bedside for body composition assessment and have evaluated its use in a racially diverse colon cancer screening subject population. An US technique using a single abdominal landmark at the left and right of the umbilicus in both males and females yielded indirect thickness estimates for SAT, DSAT and SSAT layers which were strongly associated with DXA derived estimates. DXA-derived prediction equations for total fat mass, BF%, and trunk fat mass from US estimates of SAT, SSAT and DSAT were highly correlated and without bias compared to DXA estimates in males; for females, this was also the case with the exception of BF%. In summary, the US technique to analyze abdominal adipose tissue thickness showed strong agreement compared to DXA in both males and females and generated highly comparable measurements for abdominal and total adiposity with considerable accuracy.

Declaration of competing interest

The authors declared no conflict of interest.

Acknowledgements

We would like to thank Dr. Liam McKeever, PhD, RD for reviewing this manuscript and providing expertise on prose and statistical analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2021.08.038.

Funding

This study was funded in part by the National Cancer Institute R01CA204808 (HRG, EM, LTH), 5R01CA204808-03 (SGP), 5U54MD012523-03 Sub-Project ID: 7086 (LTH, EM, HRG) and by the American Cancer Society MRSG 14-025-01-CNE (LTH).

Authors' contributions

SGP conceptualized analysis; SGP conducted statistical analysis and YZ assisted in data interpretation; EM, HRG, LTH, and JR supervised data collection; SGP, EM, and MM drafted manuscript and had primary responsibility for final content. All authors read, edited, and approved the final manuscript.

References

[1] Wang Y, Beydoun MA, Min J, Xue H, Kaminsky LA, Cheskin LJ. Has the prevalence of overweight, obesity and central obesity levelled off in the United States? Trends, patterns, disparities, and future projections for the obesity epidemic. Int J Epidemiol 2020. https://doi.org/10.1093/ije/dyz273. Epub 2020/02/06. PubMed PMID: 32016289.

[2] Hill JH, Solt C, Foster MT. Obesity associated disease risk: the role of inherent differences and location of adipose depots. Horm Mol Biol Clin Invest 2018;33(2). https://doi.org/10.1515/hmbci-2018-0012. Epub 2018/03/17. PubMed PMID: 29547393.

[3] Okorodudu DO, Jumean MF, Montori VM, Romero-Corral A, Somers VK, Erwin PJ, et al. Diagnostic performance of body mass index to identify obesity

as defined by body adiposity: a systematic review and meta-analysis. Int J Obes (Lond) 2010;34(5):791–9. https://doi.org/10.1038/ijo.2010.5. Epub 2010/02/04. PubMed PMID: 20125098.
[4] Shah M, Hurt RT, Mundi MS. Phenotypes of obesity: how it impacts management. Curr Gastroenterol Rep 2017;19(11):55. https://doi.org/10.1007/s11894-017-0598-1. Epub 2017/09/28. PubMed PMID: 28948512.
[5] Zhang M, Hu T, Zhang S, Zhou L. Associations of different adipose tissue depots with insulin resistance: a systematic review and meta-analysis of observational studies. Sci Rep 2015;5:18495. https://doi.org/10.1038/srep18495. Epub 2015/12/22. PubMed PMID: 26686961; PubMed Central PMCID: PMC4685195.
[6] Shah RV, Murthy VL, Abbasi SA, Blankstein R, Kwong RY, Goldfine AB, et al. Visceral adiposity and the risk of metabolic syndrome across body mass index: the MESA Study. JACC Cardiovasc Imaging 2014;7(12):1221–35. https://doi.org/10.1016/j.jcmg.2014.07.017. Epub 2014/12/03. PubMed PMID: 25440591; PubMed Central PMCID: PMC4268163.
[7] Walker GE, Verti B, Marzullo P, Savia G, Mencarelli M, Zurleni F, et al. Deep subcutaneous adipose tissue: a distinct abdominal adipose depot. Obesity (Silver Spring, Md) 2007;15(8):1933–43. https://doi.org/10.1038/oby.2007.231.
[8] Petrus P, Edholm D, Rosqvist F, Dahlman I, Sundbom M, Arner P, et al. Depot-specific differences in fatty acid composition and distinct associations with lipogenic gene expression in abdominal adipose tissue of obese women. Int J Obes (Lond) 2017;41(8):1295–8. https://doi.org/10.1038/ijo.2017.106. Epub 2017/05/04. PubMed PMID: 28465608; PubMed Central PMCID: PMC5550557.
[9] Marinou K, Hodson L, Vasan SK, Fielding BA, Banerjee R, Brismar K, et al. Structural and functional properties of deep abdominal subcutaneous adipose tissue explain its association with insulin resistance and cardiovascular risk in men. Diabetes Care 2014;37(3):821–9. https://doi.org/10.2337/dc13-1353. Epub 2013/11/05. PubMed PMID: 24186879.
[10] Bacchi E, Negri C, Tarperi C, Baraldo A, Faccioli N, Milanese C, et al. Relationships between cardiorespiratory fitness, metabolic control, and fat distribution in type 2 diabetes subjects. Acta Diabetol 2013. https://doi.org/10.1007/s00592-013-0519-1. Epub 2013/10/17. PubMed PMID: 24129948.
[11] Shepherd JA, Ng BK, Sommer MJ, Heymsfield SB. Body composition by DXA. Bone 2017;104:101–5. https://doi.org/10.1016/j.bone.2017.06.010. Epub 2017/06/16. PubMed PMID: 28625918.
[12] Mourtzakis M, Prado CM, Lieffers JR, Reiman T, McCargar LJ, Baracos VE. A practical and precise approach to quantification of body composition in cancer patients using computed tomography images acquired during routine care. Appl Physiol Nutr Metabol = Physiologie appliquee, nutrition et metabolisme 2008;33(5):997–1006. https://doi.org/10.1139/H08-075.
[13] Mourtzakis M, Parry S, Connolly B, Puthucheary Z. Skeletal muscle ultrasound in critical care: a tool in need of translation. Ann Am Thorac Soc 2017;14(10):1495–503. https://doi.org/10.1513/AnnalsATS.201612-967PS. Epub 2017/08/19. PubMed PMID: 28820608; PubMed Central PMCID: PMC5718569.
[14] Gallagher D, Shaheen I, Zafar K. State-of-the-art measurements in human body composition: a moving frontier of clinical importance. Int J Basic Clin Res 2008;6(4):141–8. PubMed PMID: 21234275.
[15] Teigen LM, Kuchnia AJ, Mourtzakis M, Earthman CP. The use of technology for estimating body composition (strengths and weaknesses of common modalities in a clinical setting [formula: see text]). Nutr Clin Pract 2017;32(1):20–9. https://doi.org/10.1177/0884533616676264. Epub 2016/11/12. PubMed PMID: 27834282.
[16] Sheean P, Gonzalez MC, Prado CM, McKeever L, Hall AM, Braunschweig CA. American society for parenteral and enteral nutrition clinical guidelines: the validity of body composition assessment in clinical populations. JPEN - J Parenter Enter Nutr 2019. https://doi.org/10.1002/jpen.1669. Epub 2019/06/20. PubMed PMID: 31216070.
[17] Price KL, Earthman CP. Update on body composition tools in clinical settings: computed tomography, ultrasound, and bioimpedance applications for assessment and monitoring. Eur J Clin Nutr 2019;73(2):187–93. https://doi.org/10.1038/s41430-018-0360-2. Epub 2018/11/01. PubMed PMID: 30377307.
[18] Abe T. Relationship between dual-energy X-ray absorptiometry-derived appendicular lean tissue mass and total body skeletal muscle mass estimated by ultrasound. In: Dabbs NNV, Ford M, Bass M, Loftin M, editors. International journal of clinical Medicine; 2013. p. 283–6.
[19] Paris MT, Lafleur B, Dubin JA, Mourtzakis M. Development of a bedside viable ultrasound protocol to quantify appendicular lean tissue mass. J Cachexia Sarcopenia Muscle 2017;8(5):713–26. https://doi.org/10.1002/jcsm.12213. Epub 2017/07/20. PubMed PMID: 28722298; PubMed Central PMCID: PMC5659058.
[20] Abe T, Loenneke JP, Thiebaud RS, Fujita E, Akamine T, Loftin M. Prediction and validation of DXA-derived appendicular fat-free adipose tissue by a single ultrasound image of the forearm in Japanese older adults. J Ultrasound Med 2018;37(2):347–53. https://doi.org/10.1002/jum.14343. Epub 2017/08/05. PubMed PMID: 28777477.
[21] Abe T, Fujita E, Thiebaud RS, Loenneke JP, Akamine T. Ultrasound-derived forearm muscle thickness is a powerful predictor for estimating DXA-derived appendicular lean mass in Japanese older adults. Ultrasound Med Biol 2016;42(9):2341–4. https://doi.org/10.1016/j.ultrasmedbio.2016.05.008. Epub 2016/06/21. PubMed PMID: 27321173.
[22] Clinical guidelines on the identification, evaluation, and treatment of overweight and obesity in adults: executive summary. Expert Panel on the Identification, Evaluation, and Treatment of Overweight in Adults. Am J Clin Nutr 1998;68(4):899–917. https://doi.org/10.1093/ajcn/68.4.899. Epub 1998/10/15. PubMed PMID: 9771869.
[23] Watson PF, Petrie A. Method agreement analysis: a review of correct methodology. Theriogenology Jun 2010;73(9):1167–79. https://doi.org/10.1016/j.theriogenology.2010.01.003. Epub 2010/02/09. PubMed PMID: 20138353.
[24] Lohman TG. Skinfolds and body density and their relation to body fatness: a review. Hum Biol May 1981;53(2):181–225. Epub 1981/05/01. PubMed PMID: 7239496.
[25] Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. Lancet Feb 8 1986;1(8476):307–10. Epub 1986/02/08. PubMed PMID: 2868172.
[26] Bland JM, Altman DG. Measuring agreement in method comparison studies. Stat Methods Med Res Jun 1999;8(2):135–60. https://doi.org/10.1177/096228029900800204. Epub 1999/09/29. PubMed PMID: 10501650.
[27] Faul F, Erdfelder E, Buchner A, Lang AG. Statistical power analyses using G*Power 3.1: tests for correlation and regression analyses. Behav Res Methods Nov 2009;41(4):1149–60. https://doi.org/10.3758/brm.41.4.1149.
[28] Wagner DR, Cain DL, Clark NW. Validity and reliability of A-mode ultrasound for body composition assessment of NCAA division I athletes. PLoS One 2016;11(4):e0153146. https://doi.org/10.1371/journal.pone.0153146.
[29] Pineau JC, Lalys L, Pellegrini M, Battistini NC. Body fat mass assessment: a comparison between an ultrasound-based device and a discovery A model of DXA. ISRN Obes 2013;2013:462394. https://doi.org/10.1155/2013/462394.
[30] Abe T, Tanaka F, Kawakami Y, Yoshikawa K, Fukunaga T. Total and segmental subcutaneous adipose tissue volume measured by ultrasound. Med Sci Sports Exerc Jul 1996;28(7):908–12.
[31] Paris MT, Mourtzakis M. Development of a bedside-applicable ultrasound protocol to estimate fat mass index derived from whole body dual-energy x-ray absorptiometry scans. Nutrition Jan 2019;57:225–30. https://doi.org/10.1016/j.nut.2018.04.012.
[32] Leahy S, Toomey C, McCreesh K, O'Neill C, Jakeman P. Ultrasound measurement of subcutaneous adipose tissue thickness accurately predicts total and segmental body fat of young adults. Ultrasound Med Biol Jan 2012;38(1):28–34. https://doi.org/10.1016/j.ultrasmedbio.2011.10.011.
[33] Pineau JC, Guihard-Costa AM, Bocquet M. Validation of ultrasound techniques applied to body fat measurement. Ann Nutr Metabol 2007;51(5):421–7. https://doi.org/10.1159/000111161.
[34] Goodpaster BH, Thaete FL, Kelley DE. Composition of skeletal muscle evaluated with computed tomography. Ann N Y Acad Sci May 2000;904:18–24. https://doi.org/10.1111/j.1749-6632.2000.tb06416.x.